

NAME - SAHIL JALAGAR

USN - 2VX25UE404

EXPT NO. - 2

TITLE - QUICK SORT

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In [7]: import random
import time
import matplotlib.pyplot as plt

def quick_sort(arr):
    if len(arr) <= 1:
        return arr
    pivot = arr[len(arr) // 2]
    left = [x for x in arr if x < pivot]
    middle = [x for x in arr if x == pivot]
    right = [x for x in arr if x > pivot]
    return quick_sort(left) + middle + quick_sort(right)

# Input:
arr = list(map(int, input("Enter elements separated by space: ").split()))
print("Input array: ", arr)
sorted_arr = quick_sort(arr)
print("Sorted array:", sorted_arr)

# Measure time taken to sort
start_time = time.time()
sorted_arr = quick_sort(arr)
end_time = time.time()
print("Time taken to sort:", end_time - start_time, "seconds")
print("=====")

# For n>5000
n_values = [5000, 6000, 7000, 8000]
time_values = []

for n in n_values:
    arr = [random.randint(1, 9) for _ in range(n)] # Generate random array
    start_time = time.time()
    sorted_arr = quick_sort(arr)
    end_time = time.time()
    time_taken = end_time - start_time
    print("Time taken to sort", n, "elements:", time_taken, "seconds")
    time_values.append(time_taken)

print("=====")

# Plotting the graph
plt.plot(n_values, time_values, color='red', marker='o')
plt.xlabel('Number of Elements (n)')
plt.ylabel('Time Taken (seconds)')
plt.title('Quick Sort Time Complexity Analysis')
plt.grid(True)
plt.show()
```

Input array: [60, 40, 50, 20, 10, 30, 90, 70, 80]

Sorted array: [10, 20, 30, 40, 50, 60, 70, 80, 90]

Time taken to sort: 0.00022029876708984375 seconds

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Time taken to sort 5000 elements: 0.006892204284667969 seconds
Time taken to sort 6000 elements: 0.009347200393676758 seconds
Time taken to sort 7000 elements: 0.008056879043579102 seconds
Time taken to sort 8000 elements: 0.01140141487121582 seconds
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